

Felipe Pérez Silva

Ph.D. Student in Mathematics, George Mason University

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Research Interests

Nonlinear analysis of partial differential equations: existence and multiplicity of solutions for quasilinear elliptic problems, using variational techniques such as the Mountain Pass theorem and the method of sub- and super-solutions.

Numerical analysis of partial differential equations: numerical methods for constrained PDE problems via Fenchel duality theory, with guaranteed a posteriori error control.

Education

2024 – present	Ph.D. in Mathematics <i>George Mason University</i> , Fairfax, VA, USA Advisor: Dr. Harbir Antil
2017 – 2023	Professional Degree in Mathematical Engineering (<i>Ingeniería Civil Matemática</i>) <i>Universidad Técnica Federico Santa María</i> , Valparaíso, Chile Six-year integrated program. B.Sc. in Mathematical Engineering awarded in 2020 (years 1–4). The M.Sc. below was pursued concurrently during years 5–6, with a single joint thesis defense.
2021 – 2023	M.Sc. in Mathematics <i>Universidad Técnica Federico Santa María</i> , Valparaíso, Chile Advisor: Dr. Leonelo Iturriaga <i>Thesis (joint defense with Mathematical Engineering):</i> “Existence and multiplicity of positive solutions for a quasilinear problem with nonhomogeneous Dirichlet condition.”

Publications

Submitted / Under Review

1. H. Antil, R. Löhner, and **F. Pérez**, “Optimal boundary control of diffusion on graphs via linear programming,” *Submitted, 2025*. [arXiv:2511.03129](https://arxiv.org/abs/2511.03129).
2. L. Iturriaga and **F. Pérez**, “Existence and multiplicity of positive solutions for a quasilinear problem with nonhomogeneous Dirichlet condition,” *Submitted, 2025*.

In Preparation

3. H. Antil, A. Kaltenbach, **F. Pérez**, T. Prieto, and V. Lagos, “Convex duality and guaranteed a posteriori error control for curl–curl Maxwell problems with pointwise field bounds,” *In preparation, 2026*.
4. H. Antil, K. L. A. Kirk, and **F. Pérez**, “Duality framework for flux constrained flow: analysis and numerics,” *In preparation, 2026*.
5. H. Antil and **F. Pérez**, “Duality framework for the double obstacle problem: analysis and numerics,” *In preparation, 2026*.

Research Experience

2024 – present	Graduate Research Assistant <i>George Mason University</i> · PI: Dr. Harbir Antil Research on convex-duality methods and a posteriori error control for constrained PDE problems, including obstacle problems, flux-constrained flows, and curl–curl Maxwell systems with pointwise field bounds. Optimal boundary control of diffusion on graphs via linear programming.
2021 – 2023	Research Assistant <i>Universidad Técnica Federico Santa María</i> · PI: Dr. Leonelo Iturriaga Research on existence and multiplicity of positive solutions for quasilinear elliptic problems involving the p -Laplacian under nonhomogeneous Dirichlet conditions. Variational and sub/super-solution methods.
Summer 2020	Software Engineering Intern <i>Inria Chile</i> , Santiago, Chile Optimized Django REST Framework backend queries to reduce database load. Implemented React and RxJS frontend improvements for application performance.

Teaching Experience

Instructor of Record

Spring 2024	<i>Universidad Técnica Federico Santa María</i> Term Professor (sole instructor): Analytic Geometry; Calculus I; Calculus II.
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Graduate Teaching Assistant

2024 – present	<i>George Mason University</i> Calculus I and Calculus II: led recitation sections, held office hours, graded exams and assignments.
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Laboratory & Teaching Assistant

2023	<i>Universidad Técnica Federico Santa María</i> Laboratory Assistant: Differential Calculus; Algebra & Geometry.
2018 – 2022	<i>Universidad Técnica Federico Santa María</i> Teaching Assistant for Differential Calculus, Integral Calculus, Algebra & Geometry, Linear Algebra, and Partial Differential Equations.

Talks & Presentations

2025	Finite Element Circus , George Mason University. <i>Duality Framework for Flux Constrained Flow: Analysis and Numerics.</i>
2025	Seminar on Calculus of Variations , Georgetown University. Invited speaker.
2024	Seminar on Control of Partial Differential Equations , George Mason University. Presenter and attendee; talks on the Fenchel conjugate and primal–dual formulations for PDE-constrained problems.

2021 **Seminar on Partial Differential Equations** (Doctoral Program), Universidad Técnica Federico Santa María. Presenter on minimax methods (Mountain Pass and linking theorems), the Nemytskii operator, and the deformation lemma.

Workshops & Posters

2026 **ICERM**, Brown University. *Simulation-Based Optimization with Applications*. Poster presentation.

2025 **IMSI**, University of Chicago. *Optimal Control and Decision Making Under Uncertainty for Digital Twins*. Poster presentation.

Skills

Programming	Python (NumPy, SciPy), MATLAB, C++, $\LaTeX$.
Numerical PDE	FEniCS, NGSolve.
Web (legacy)	Django REST Framework, React, RxJS (from Inria internship).
Languages	Spanish (native); English (professional working proficiency).